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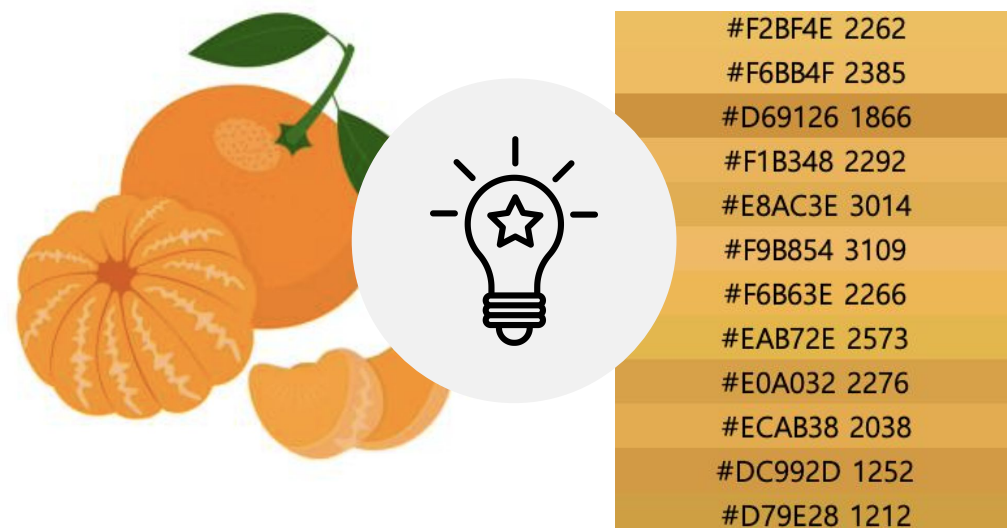
Development of AI Sugar Content Prediction

Sorting Device Using Citrus Peel Color

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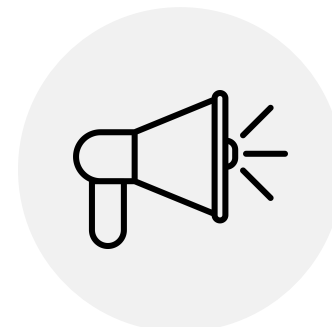
Purpose of Design



- Citrus with a deep reddish-orange color is considered delicious by consumers.
- Can an AI analyze the taste of a citrus by its color?



- What are the RGB and HSV distribution of high-sugar citrus?
- If you can extract a citrus from an image taken by the sorting device, remove the background, and extract the color, wouldn't it be possible to get the RGB and HSV of the citrus?



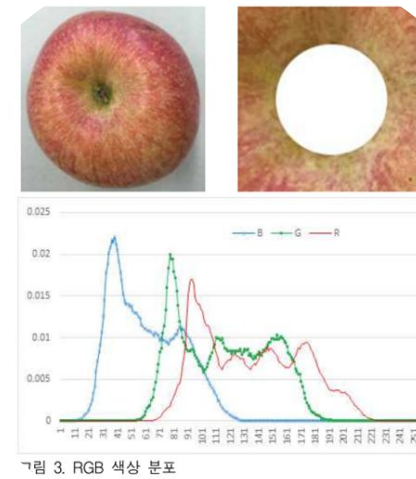
- Predict the sugar content of citrus, a local specialty, based on a paper that analyzed correlation between apple peel color and sugar content.
- Develop a sorting device to sort citrus by sugar content

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Ideas for Work



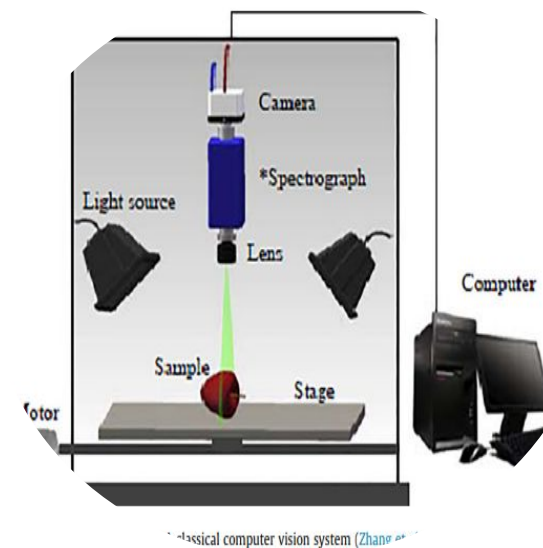
Implement a citrus sugar content prediction model based on citrus peel color and develop an AI sugar content prediction sorting device.



Analysis of Apple Colors and Sugar Contents Using Linear Regression

A paper analyzing the association between color and sugar content in apples from 2020 to 2022.

A paper analyzing the association between RGB, HSV, and Lab colors and sugar content of apples.



Fruits and vegetables quality evaluation using computer vision: A review

An analysis of the quality assessment of fruits based on color, texture, size, shape, and defects in 2018.

Design Process

Process 1

Build a citrus sorting device

Process 2

1. Build a sugar content prediction model using citrus peel color
2. Implement classification of citrus by sugar content using a sugar content prediction model

Process 3

Develop AI sugar content prediction sorting device

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Build a Citrus Sorting Device

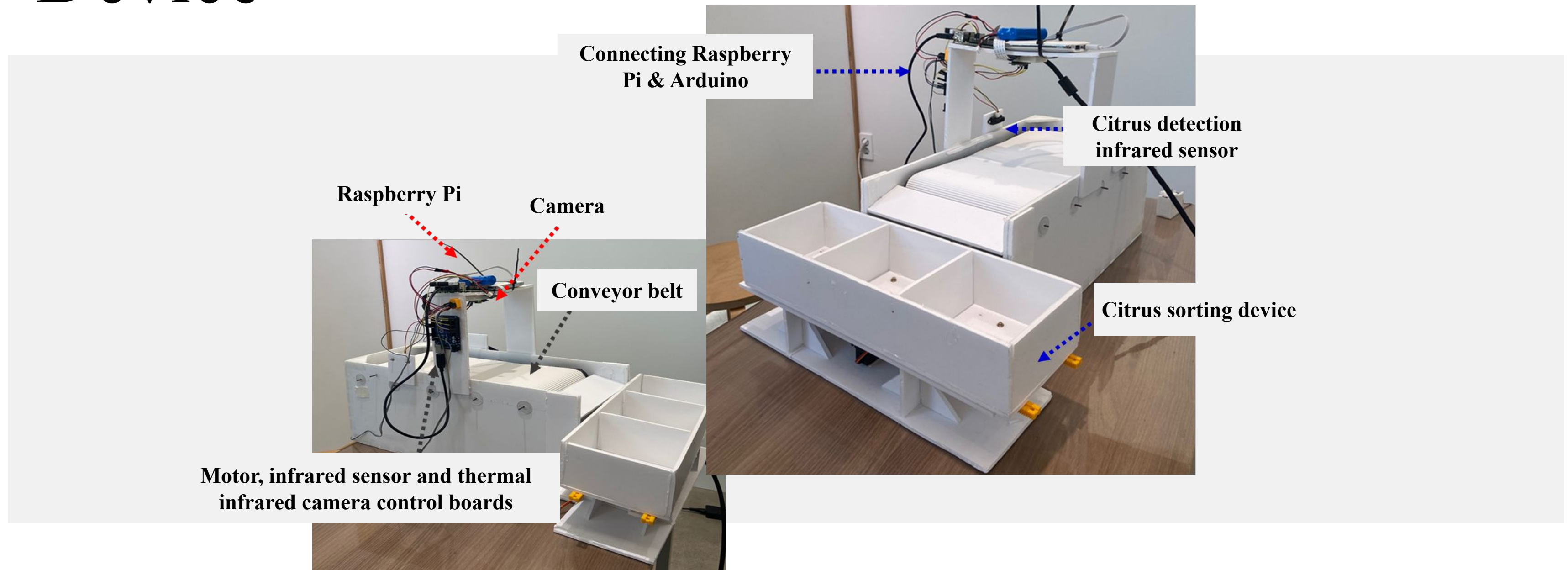
Citrus detection : Infrared sensor (detects citrus on conveyor belt)

Citrus conveyor belt : 360-degree servo motor (controls connecting the column to the motor)

Capturing citrus peel color : Raspberry Camera (take a picture and send it to the server)

Citrus sorting device control board : Arduino board

Citrus sorting device : Servomotors and rails (Citrus sorting device control)



Sugar Content Prediction Model

Citrus object detection [Citrus coordinate extraction] :

Using YOLO as an image-based object detection algorithm, YOLO predicts bounding box and class accuracy in an image, detects a citrus in the image and extracts the coordinates of the citrus.

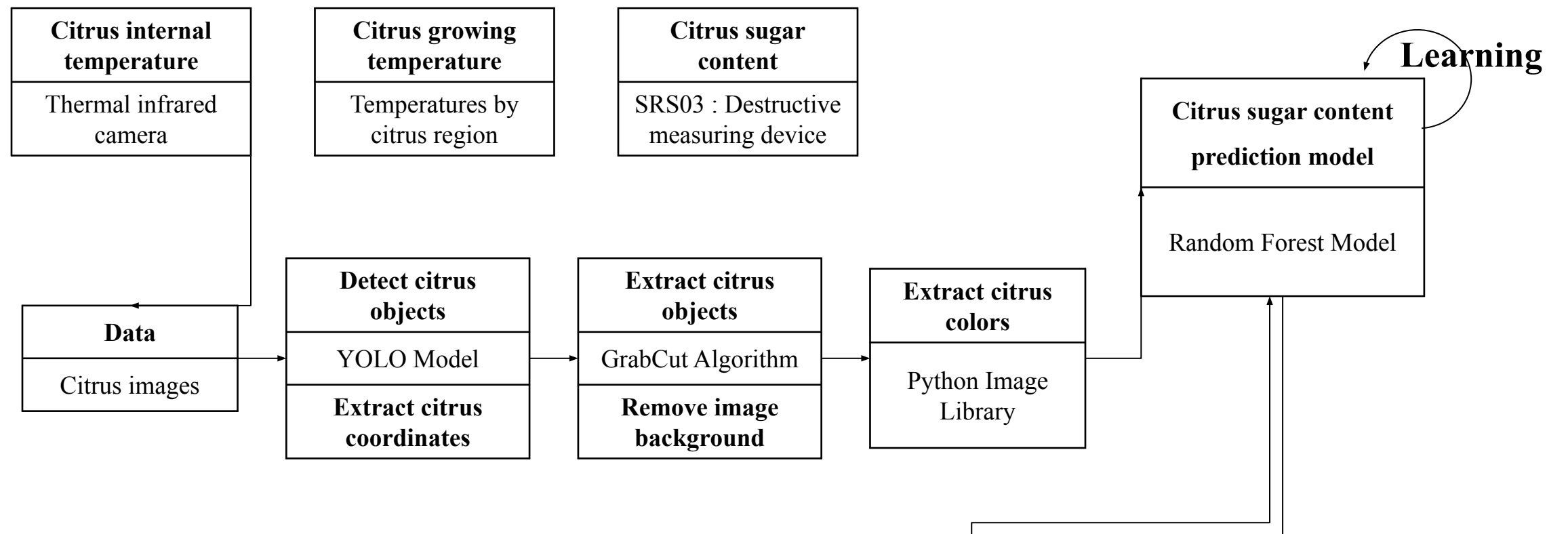
Extract citrus image [remove image background] :

Using the GrabCut algorithm, an image extraction algorithm for the foreground of an image. square the area containing the foreground of the image, mark the background and missing foreground parts, and re-extract the image.

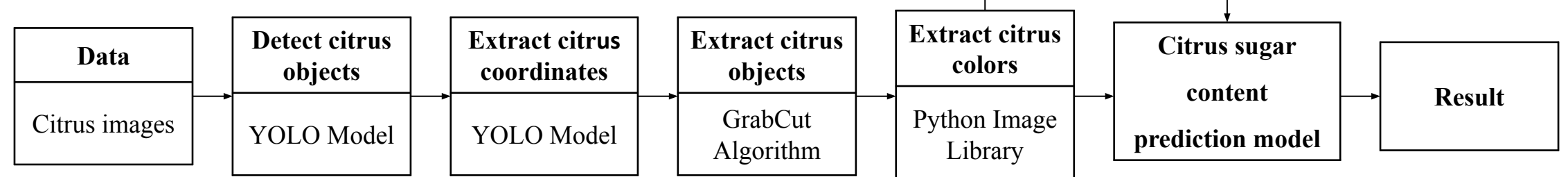
Extracting the color of a citrus : Extracting the color of a citrus using the Python Image Model (Pillow), a library for extracting colors from images in Python

Implement a citrus sugar prediction model : Using Random Forest Model, an ensemble learning method used for classification and regression in machine learning, Random Forest Model can identify the importance of each variable, has high accuracy even with many missing and outliers, and is not prone to overfitting

Stages of learning

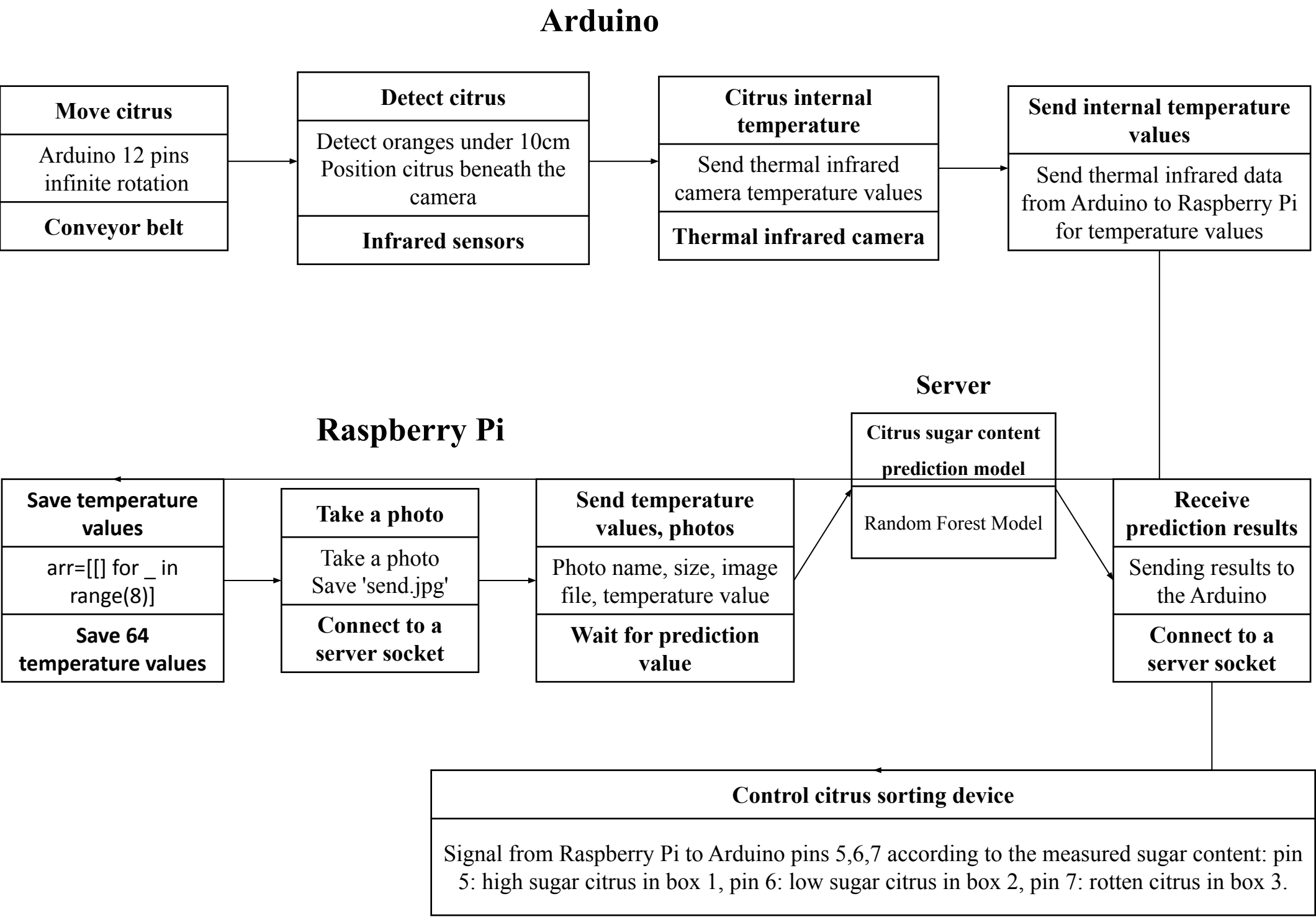
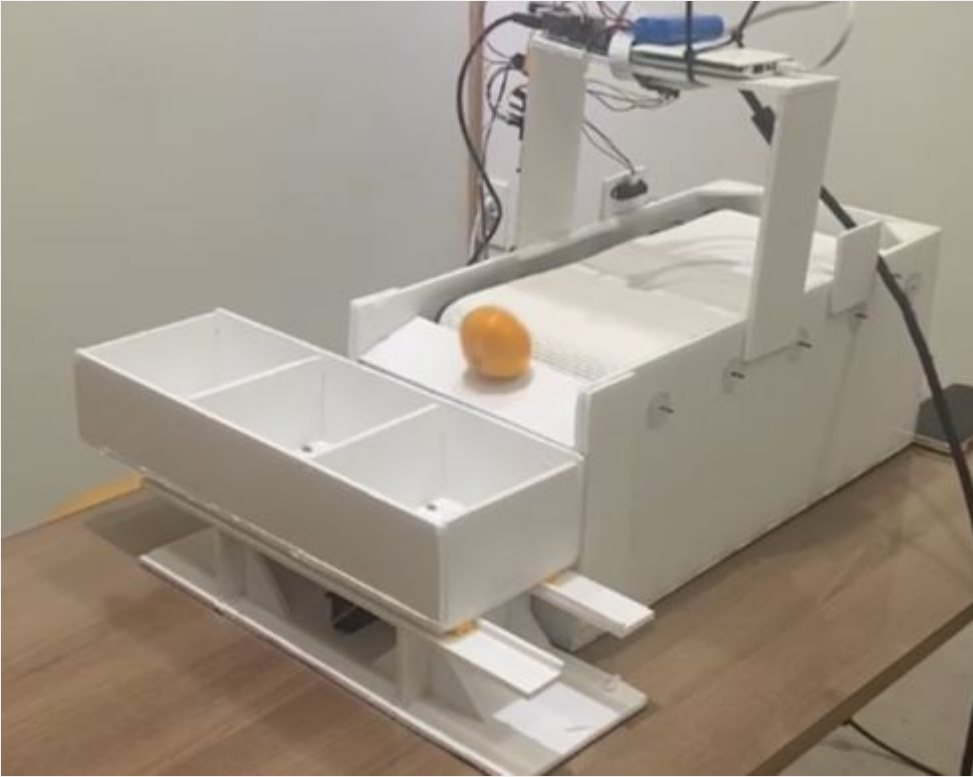


Stages of prediction



AI Sugar Content Prediction

Sorting Device

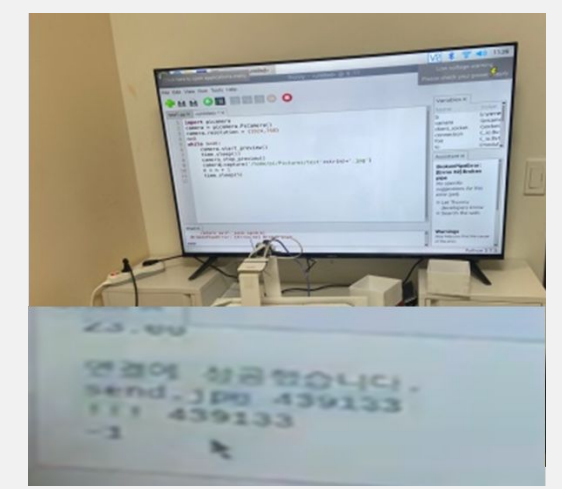


08 Operation Progress of Sugar Content Prediction Sorting Device



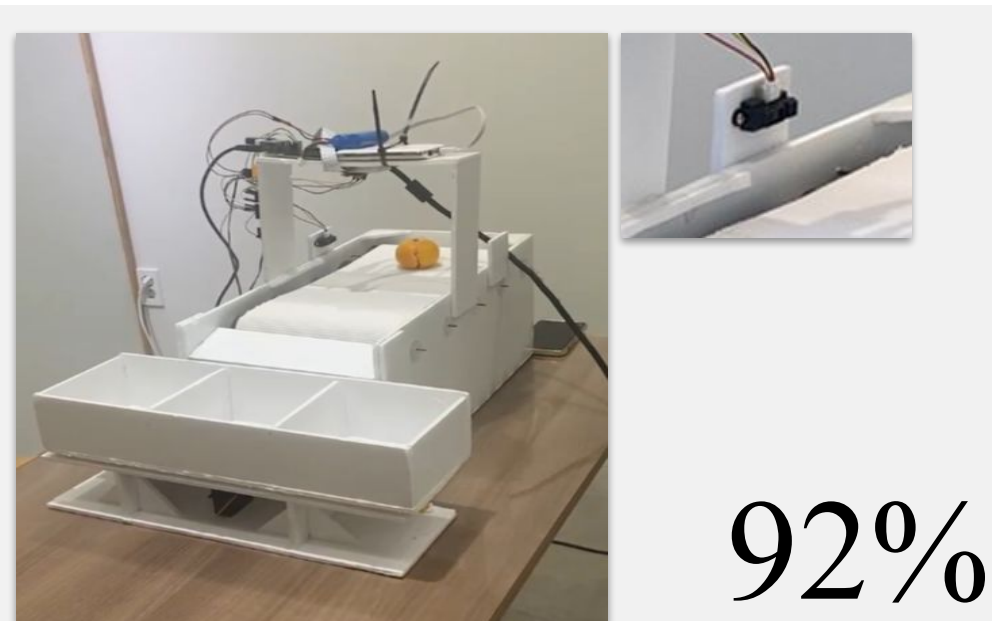
1. Place the citrus on the conveyor belt.
2. When the infrared sensor detects a citrus, the conveyor belt stops after 2 seconds

3. After the Arduino thermal infrared camera is operational, send the values to the Raspberry Pi.
4. The camera connected to the Raspberry Pi takes a picture of the citrus and sends the file to the server computer.



5. The server uses the learning model to send one of three values to the citrus sorting device: high sugar (1), low sugar (0), or bad (-1).
6. Upon receiving this signal, the conveyor belt will start moving again.
7. The citrus sorter moves according to the values 1, 0, and -1 to place the citrus in the sorting bin.

09 Citrus Sorting Device Operation Experiment



92%

Citrus Detection

Using infrared sensors

- Using infrared sensors to detect citrus moving on a conveyor belt

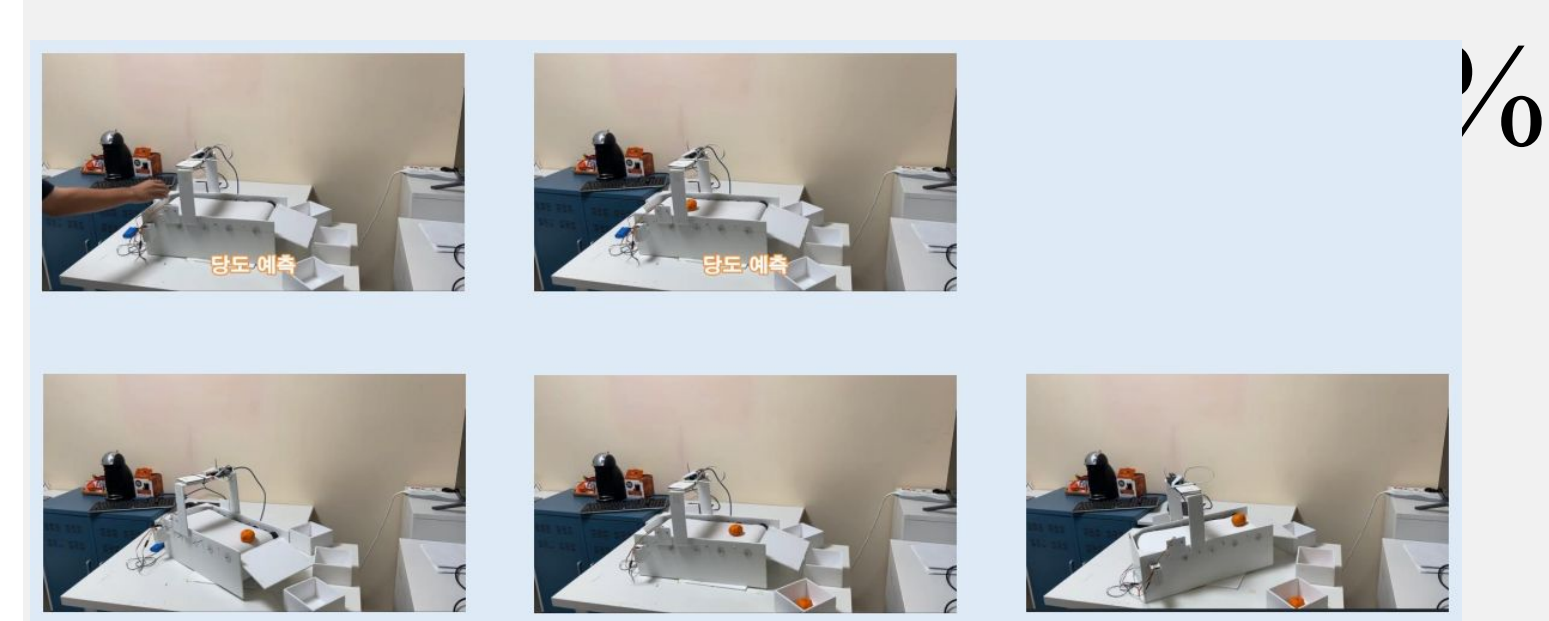


90%

Raspberry Camera

Capture citrus and sending them to a server

- Using a camera to capture the citrus and extract the color of the peel to use as training data to predict the sugar content of citrus.



Citrus Classification Experiment

Classify sugar content after receiving data from the server

- Sort citrus into three categories: high-sugar, low-sugar, and bad, using motors and rails based on the values passed to them.

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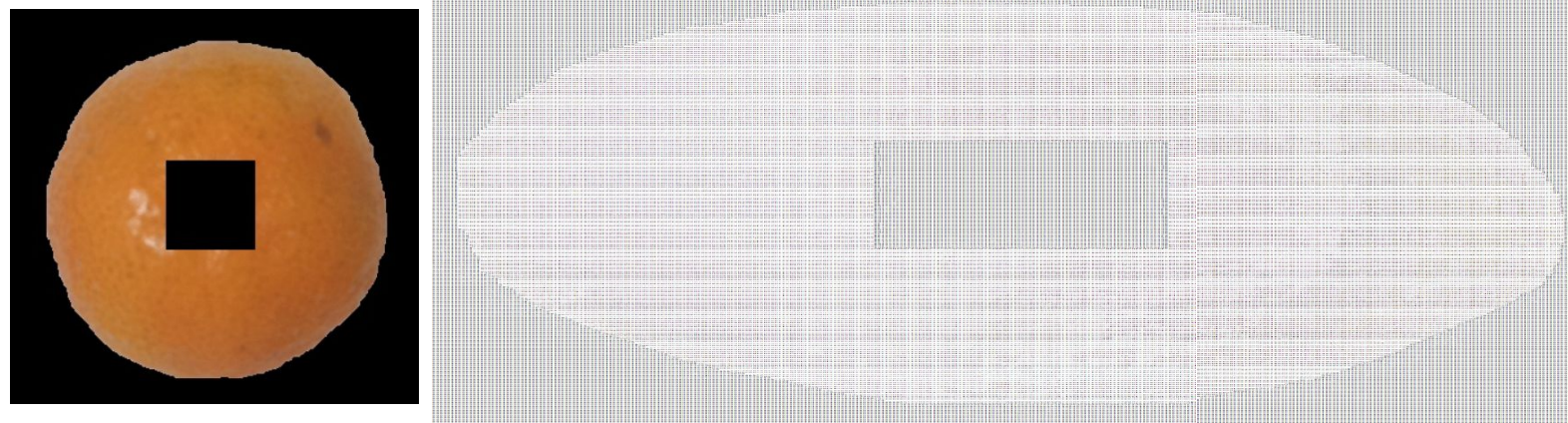
Can we use the color of a citrus peel to analyze its sugar content?

Analyze correlation between RGB, HVS of captured image and citrus sugar content

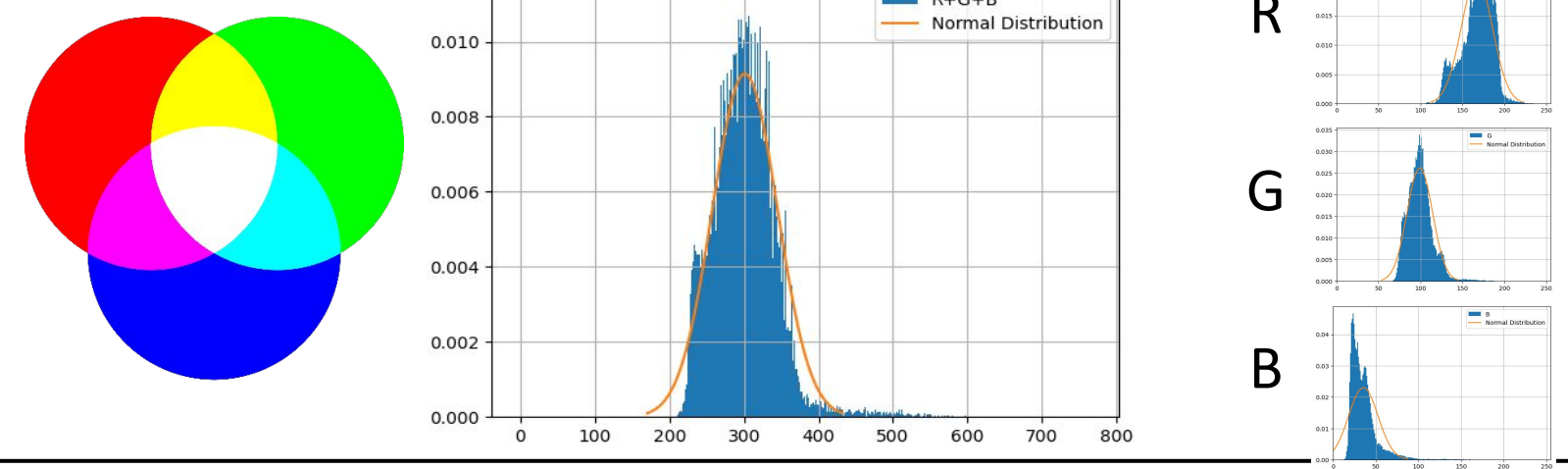
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11 Extracting Color from a Citrus Image

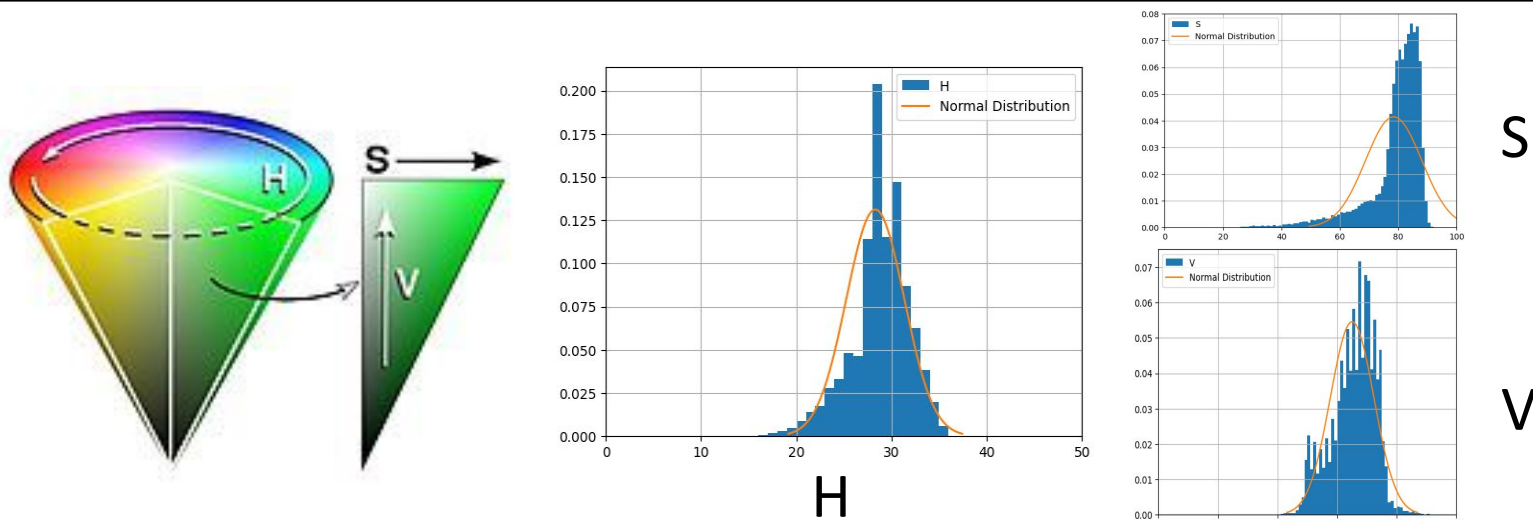
Extract the color of a citrus image [in 1px increments].



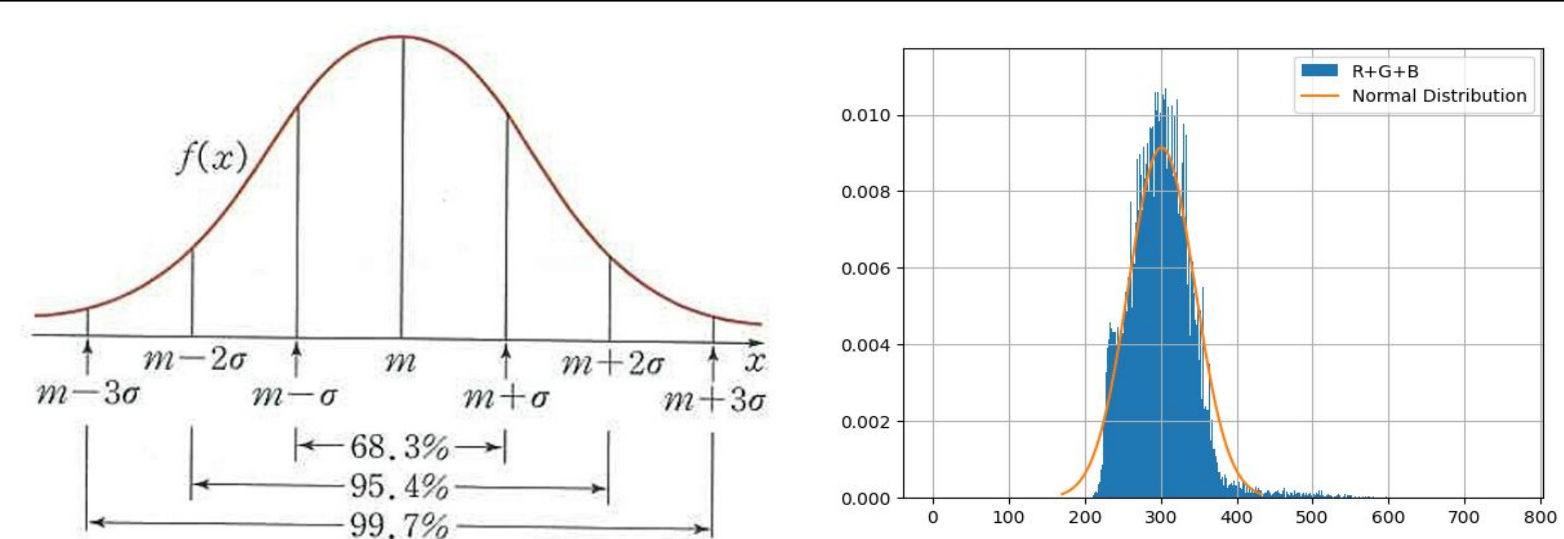
R, G, and B values expressed as 0-255



Expressed as H (hue 0-360), S (saturation), and V (brightness) values

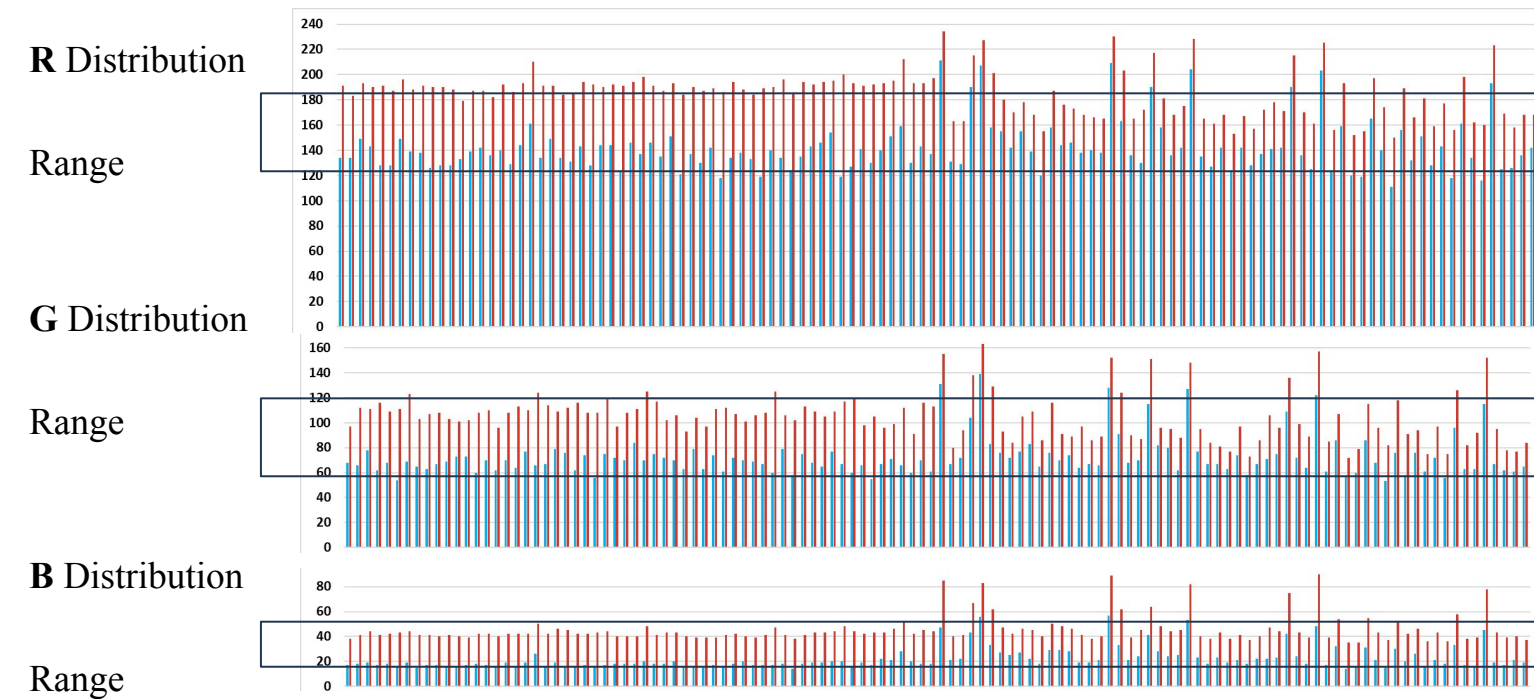


Analyze the relationship between sugar content using RGB normal distribution



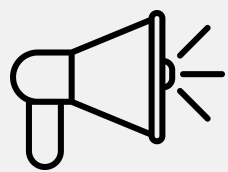
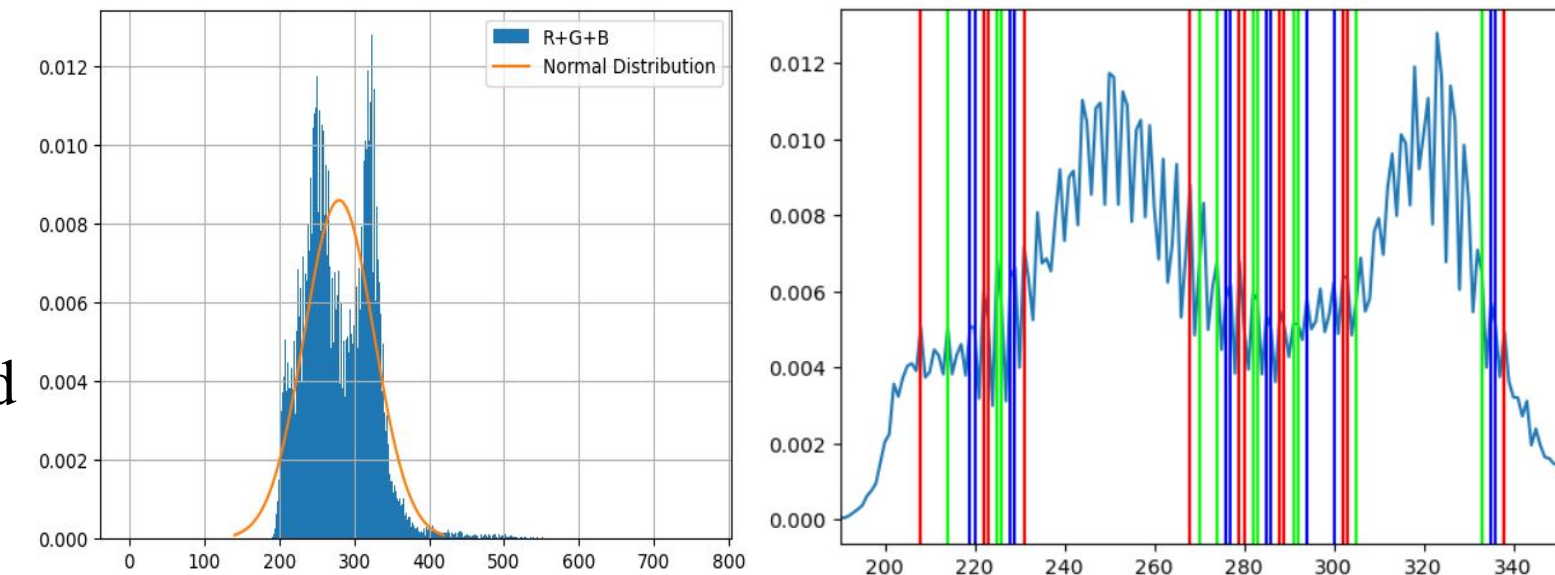
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Analysis and Conclusions



- R, G, and B distribution ranges for all data
- R, G, and B has similar distribution ranges
- Represent R,G,B as a combined value
- Relation between the distribution of R+G+B and sugar content

- Problems with the R+B+G normal distribution
- Existence of shapes that do not follow a normal distribution graph
- Extract the 80% of values that are highly distributed
- Extract data from a range of start and end



Analyze correlation using citrus shipped from a certain region, at a certain time of year.

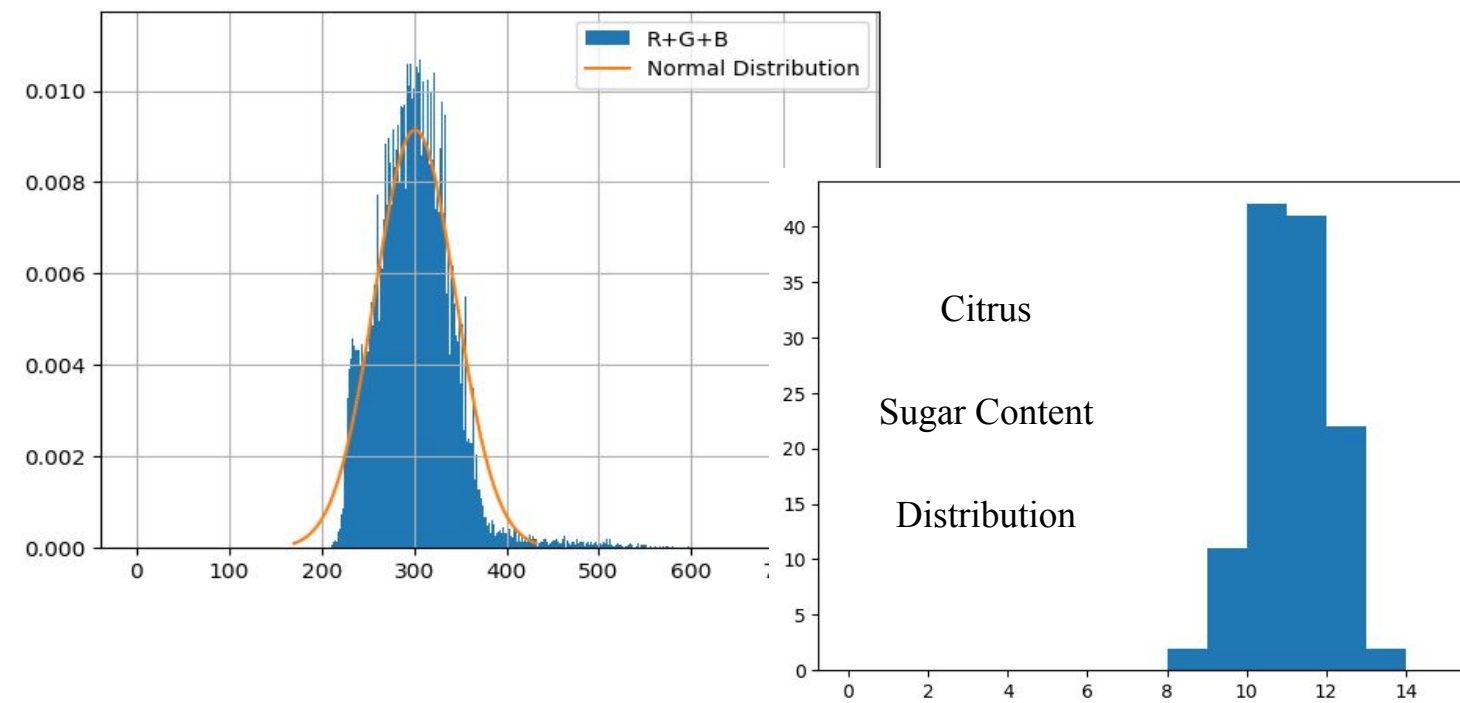
Can predict sugar content using a model trained on the color of citrus peels and sugar content!

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Sugar Content Prediction Accuracy

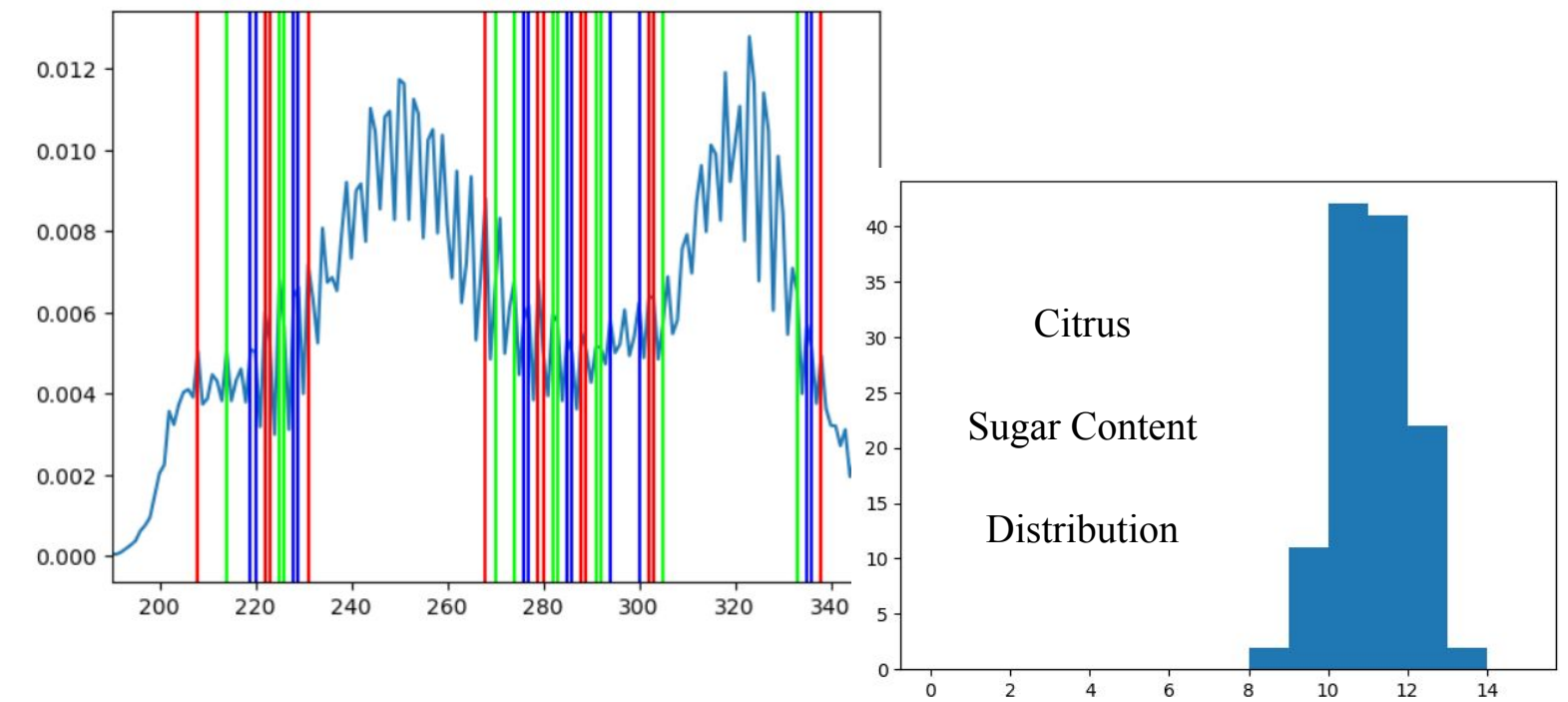
67%

- 120 data [100 training, 20 testing].
- Analyze the correlation between sugar content of the citrus and $-\sigma$ to σ values in the normal distribution of R+G+B



76%

- Extract the 80% range with a large distribution of R+G+B
- Analyze the relationship with sugar content



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Thank you!

Do you have any questions?